PlaceMate: Placement Management Portal

Project Report and Comprehensive System Documentation

A Unified Web-Based Campus Placement Automation Platform
Academic Project Documentation | Multi-Role Enterprise Architecture

Index / Table of Contents

Sr. No.	Topic Name	Page No.
1	**1. Introduction**	**1**
2	**2. Project Purpose and Objectives**	**1**
3	**3. Technological Stack**	**2**
4	**4. Database Architecture**	**2**
5	**5. Application Workflow and Screenshots**	**4**
	5.1 Common Side: Authentication & Access Control	4
	5.2 Student Side: Career Portal & Applications	6
	5.3 Company Side: Recruiter & Placement Management	12
	5.4 Admin Side: Placement Officer Administration	21
	5.5 Automated Email Notification System	25
6	**6. Conclusion**	**28**

** Note: In accordance with academic guidelines, document page numbering commences immediately following the Index table.*

1. Introduction

Campus recruitment represents one of the most vital institutional events in any higher education college or university. Annually, hundreds of students compete for career opportunities published by visiting corporate employers. However, in the vast majority of institutions, this multifaceted recruitment workflow remains managed through disparate, manual channels—including spreadsheets, Google Forms, email distribution lists, and informal instant messaging channels.

This conventional paradigm creates severe administrative friction: students frequently overlook critical drive deadlines, placement officers dedicate days to manual academic verification (cross-referencing minimum CGPA thresholds, active backlogs, and departmental eligibility), and recruiters encounter disorganized application spreadsheets without standardized resume access or structured interview pipelines.

PlaceMate is an enterprise-grade, web-based campus placement automation portal engineered to resolve these challenges comprehensively. By establishing three distinct, role-based environments—the Student Portal, the Company Portal, and the Placement Officer (Admin) Portal—PlaceMate digitizes the end-to-end recruitment lifecycle. The system centralizes student portfolios, automates real-time academic eligibility evaluations, manages multi-stage job applications, facilitates interview scheduling with automated conflict handling, sends branded transactional emails, and visualizes institutional recruitment analytics.

2. Project Purpose and Objectives

Project Purpose

The primary purpose of PlaceMate is to eliminate manual placement redundancies by delivering a unified, verifiable, and transparent digital campus recruitment platform. Serving as an institutional single source of truth, PlaceMate guarantees that students view and apply strictly to drives for which they satisfy academic requirements, visiting corporate recruiters configure targeted drives and manage candidate evaluation stages efficiently, and college administrators maintain institutional oversight over placement rates and hiring outcomes.

Core Objectives

- **Role-Based Portals:** To provide dedicated, cryptographically protected portals for Students, Corporate Recruiters, and Placement Officers with granular access controls.
- **OTP-Verified Authentication:** To validate official institutional email addresses during registration via time-limited 6-digit OTP delivery through SMTP, eliminating fraudulent registrations.
- **Automated Eligibility Filtering:** To evaluate student academic criteria (CGPA, active backlogs, graduation year, and degree department) against corporate drive cutoffs in real time prior to application submission.
- **Real-Time Application Tracking:** To furnish students with transparent, instantaneous tracking across recruitment stages: Applied, Shortlisted, Interview Scheduled, Selected, and Rejected.
- **Interview Scheduling & Reschedule Engine:** To enable recruiters to schedule interviews with direct virtual meeting integration (Google Meet, Teams, Zoom), while empowering students to request formal reschedules with conflict reasons.

- **Automated Email Notification System:** To automatically dispatch responsive, professional HTML notifications upon key milestones (account verification, drive application receipt, shortlisting, interview invitations, and final job offer selections).
- **Interactive Visual Analytics:** To deliver dynamic graphical charts illustrating departmental placement rates, salary package distributions, and recruiter hiring funnels via Chart.js.
- **Modern Responsive UI & Accessibility:** To provide an aesthetically polished user interface with zero-flicker light and dark mode toggling, fluid layout responsiveness, and accessibility compliance.

3. Technological Stack

PlaceMate is engineered using a resilient, production-ready full-stack architecture selected for security, high performance, and rapid developer velocity:

System Layer	Technology	Functional Implementation
Backend Framework	**Python 3.13 & Django**	Executes business logic, request routing, ORM database transactions, user session management, and robust built-in security protections against CSRF, SQL injection, and XSS.
Database Layer	**PostgreSQL (Supabase)**	Maintains relational integrity, structured candidate profiles, placement drives, and application records with strict foreign key constraints and transactional ACID guarantees.
Frontend Architecture	**HTML5 & Vanilla CSS**	Developed with modular CSS design tokens, modern flexbox and grid layouts, accessible UI components, and integrated light/dark theming without heavy external CSS dependencies.
Client-Side Logic	**Vanilla JavaScript (ES6+)**	Powers interactive modal interfaces, client-side form validation, real-time OTP countdown timers, dynamic theme toggling, and AJAX asynchronous actions.
Data Visualization	**Chart.js**	Generates interactive doughnut charts, bar charts, and recruitment funnel graphs for student academic insights and corporate talent analytics.
Email Services	**Django SMTP & Gmail API**	Delivers automated, responsive HTML email notifications for verification codes, calendar invites, status milestones, and secure password recovery.
Static Asset Engine	**WhiteNoise**	Provides production-grade compression, caching, and serving of static assets and stylesheets directly from the web application server.

4. Database Architecture

The database architecture of PlaceMate follows normalized relational modeling principles mapped via Django's Object-Relational Mapping (ORM). The schema guarantees separation of concerns across user roles while maintaining foreign key integrity and audit trails.

Core Relational Entities

- **User (Custom Authentication Model):** Inherits from Django's AbstractUser. Uses email as the unique primary login identifier and incorporates role attributes ('student', 'company', 'officer') and email_verified validation flags.
- **StudentProfile:** Maintains a 1-to-1 relationship with User. Stores academic metrics including full candidate name, institutional roll number, department (Foreign Key), semester, CGPA, graduation passing year, active backlogs, profile verification flag, and resume file upload.
- **CompanyProfile:** Maintains a 1-to-1 relationship with User. Contains corporate entity metadata including company name, corporate logo, industry vertical, official website URL, headquarters address, description, and admin approval status.
- **OfficerProfile:** Maintains a 1-to-1 relationship with User. Represents college placement cell directors who administer campus recruitment drives and verify corporate partners.
- **Department and Skill:** Master lookup models providing institutional standardization for college branches (e.g., Computer Engineering, Information Technology) and professional competencies (e.g., Python, Django, React, SQL).
- **PlacementDrive:** Created by CompanyProfile. Encapsulates job opening specifications: job role designation, compensation package (CTC in LPA), job location, eligible departments (Many-to-Many), required skills (Many-to-Many), minimum CGPA cutoff, maximum permitted backlogs, target passing year, application deadline, and lifecycle status ('draft', 'published', 'closed').
- **Application:** Junction model connecting StudentProfile and PlacementDrive. Records candidate application state ('applied', 'shortlisted', 'interview_scheduled', 'selected', 'rejected'), interview meeting URLs, reschedule conflict notes, and audit timestamps.

Entity Relationship Overview

Source Entity	Relationship	Target Entity	Operational Scope
User	**One-to-One**	**StudentProfile**	Each candidate account maps uniquely to an academic student profile.
User	**One-to-One**	**CompanyProfile**	Each corporate recruiter account maps to an organizational profile.
User	**One-to-One**	**OfficerProfile**	Each administrative coordinator maps to a placement staff profile.
StudentProfile	**Many-to-One**	**Department**	Every student belongs to an academic college branch.
StudentProfile	**Many-to-Many**	**Skill**	Students tag multiple verified technical competencies in their profile.
CompanyProfile	**One-to-Many**	**PlacementDrive**	A registered company can publish multiple campus placement drives.
PlacementDrive	**Many-to-Many**	**Department**	Recruitment drives define one or more eligible student departments.
StudentProfile	**One-to-Many**	**Application**	Candidates can submit multiple applications across eligible drives.
PlacementDrive	**One-to-Many**	**Application**	A published drive receives and tracks

applications from multiple candidates.

5. Application Workflow and Screenshots

This section provides a rigorous visual demonstration of PlaceMate's user interface and functional workflows across all operational portals: Common Side (Authentication), Student Side, Company Side, Admin Side (Placement Officer Portal), and the Automated Email Notification System. All 50 system figures are presented with technical descriptions illustrating real-world operation.

5.1 Common Side: Authentication and Access Control

PlaceMate implements role-segregated onboarding and self-service account management. Users verify email identity through time-limited one-time passwords (OTP) and securely authenticate with credentials.

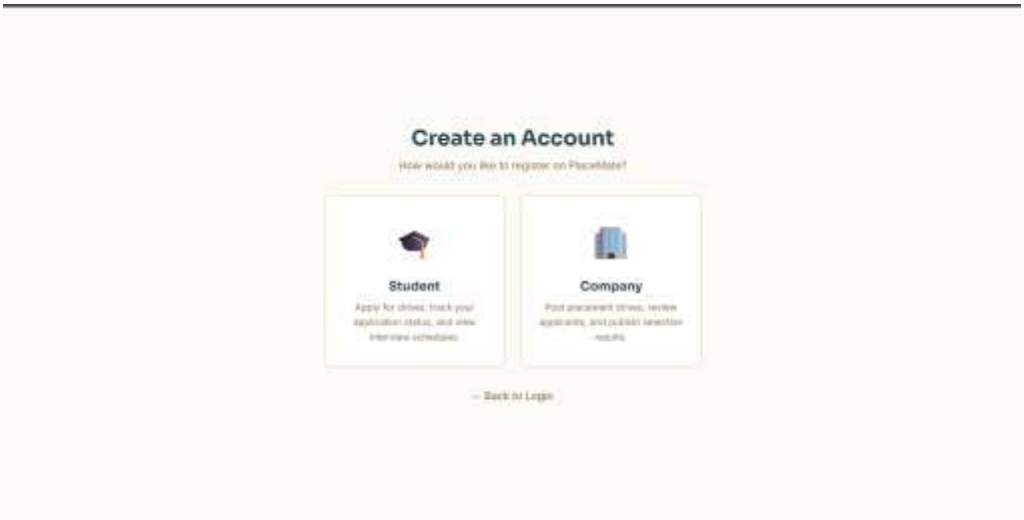


Figure 1: Registration Role Selection Screen (Common Side)

The central onboarding gateway allowing prospective users to select their intended portal role: registering as an aspiring student seeking campus opportunities or as a corporate recruitment partner looking to hire university talent.

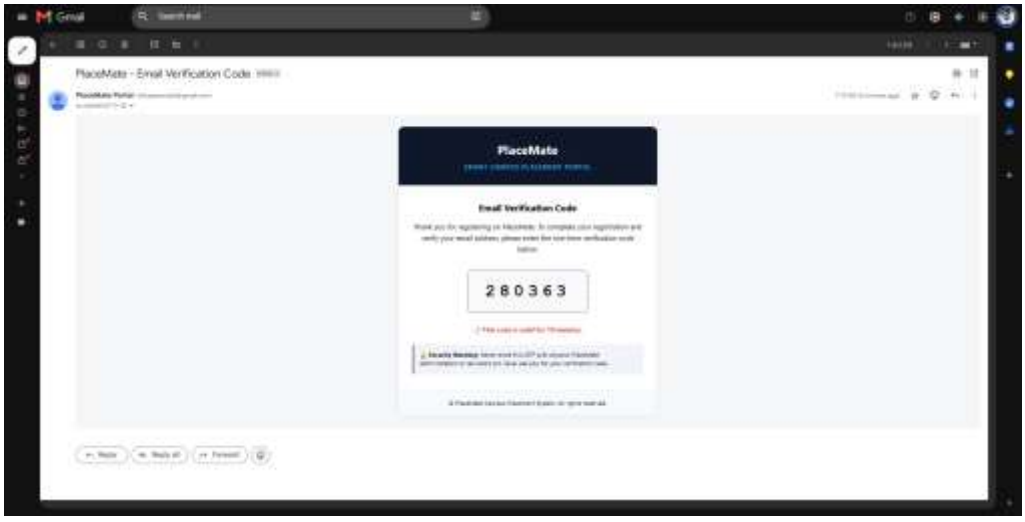


Figure 2: Real-Time Email OTP Verification Screen (Common Side)

The secure 6-digit OTP verification dialog integrated into the registration workflow. It enforces two-factor identity verification via SMTP email delivery, preventing unauthorized or automated account registrations.

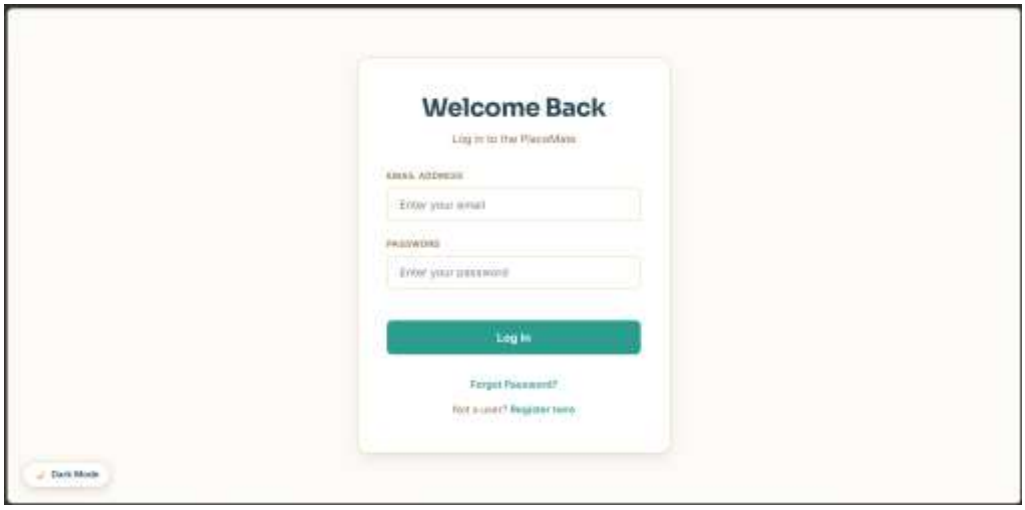


Figure 3: PlaceMate Central Login Screen (Common Side)

The unified authentication portal supporting multi-role sign-in with email and password credentials, automatic role-based dashboard redirection, and theme toggling.

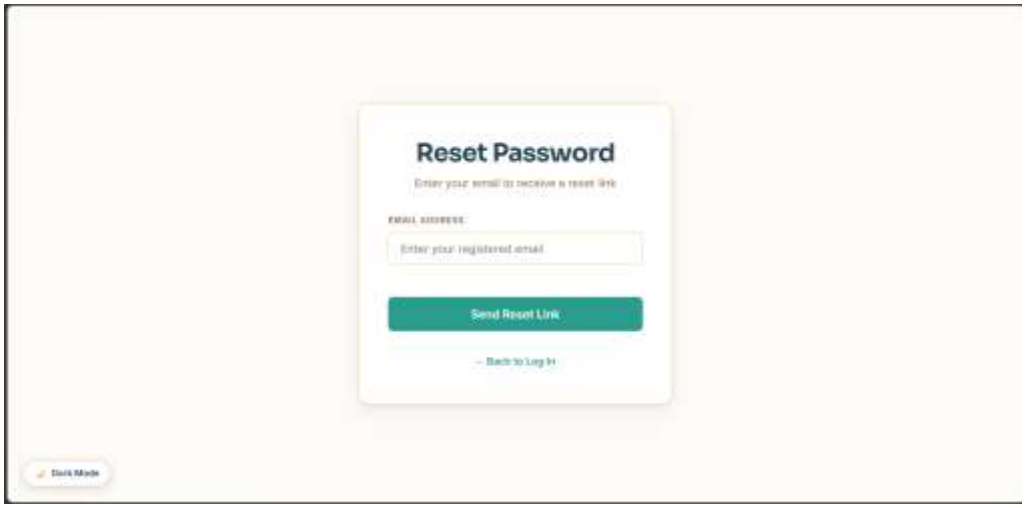


Figure 4: Password Reset Request Screen (Common Side)

Self-service password recovery interface allowing users to enter their registered email address to receive a secure, time-limited, cryptographically signed password reset link.

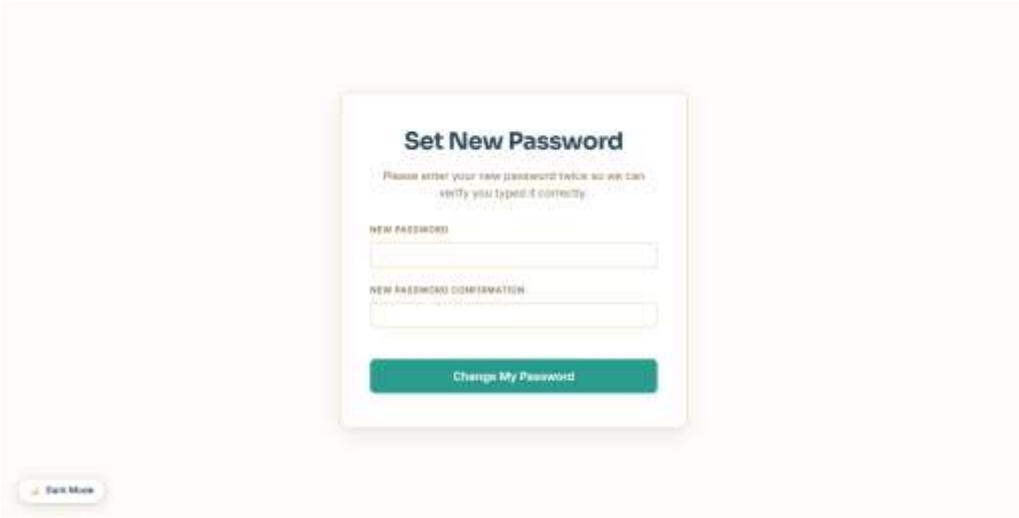


Figure 5: Set New Password Screen (Common Side)

Secure token-authenticated password reset page enforcing institutional password complexity standards including minimum length, character diversity, and confirmation validation.

5.2 Student Side: Career Portal and Applications

The Student Portal acts as a comprehensive career workstation enabling candidates to maintain their academic qualifications, discover campus recruitment drives, verify cutoffs automatically, track recruitment stages, request interview reschedules, and analyze placement outcomes.



Figure 6: Student Account Registration Form (Student Side)

Comprehensive candidate signup form collecting personal credentials, institutional roll number/USN, registered email, academic department, and graduation year with integrated client-side validation.

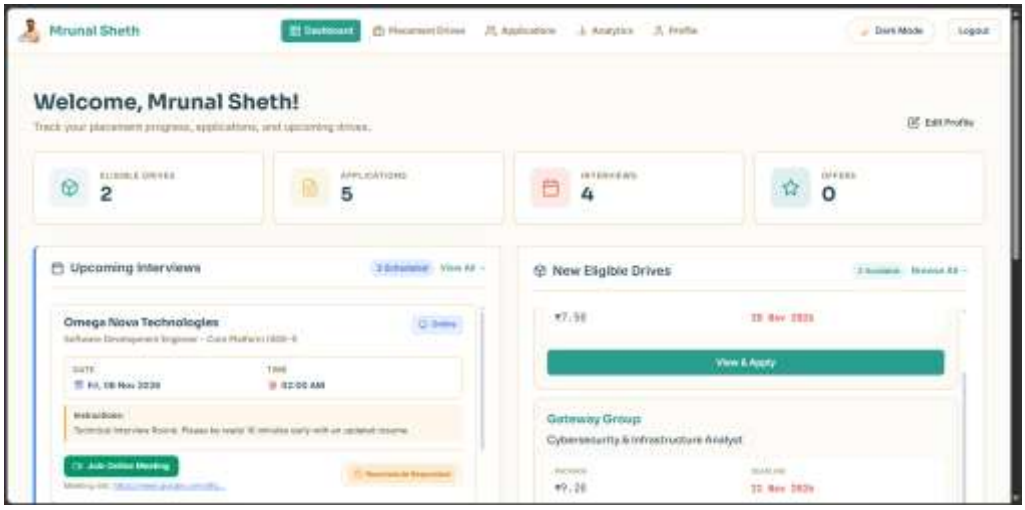


Figure 7: Student Dashboard - Overview & Quick Stats (Student Side)

Upper section of the Student Dashboard showcasing candidate welcome greetings, key recruitment summary metrics (eligible drives, submitted applications, shortlisted status), and upcoming interview notifications with direct 'Join Meeting' and 'Request Reschedule' controls.

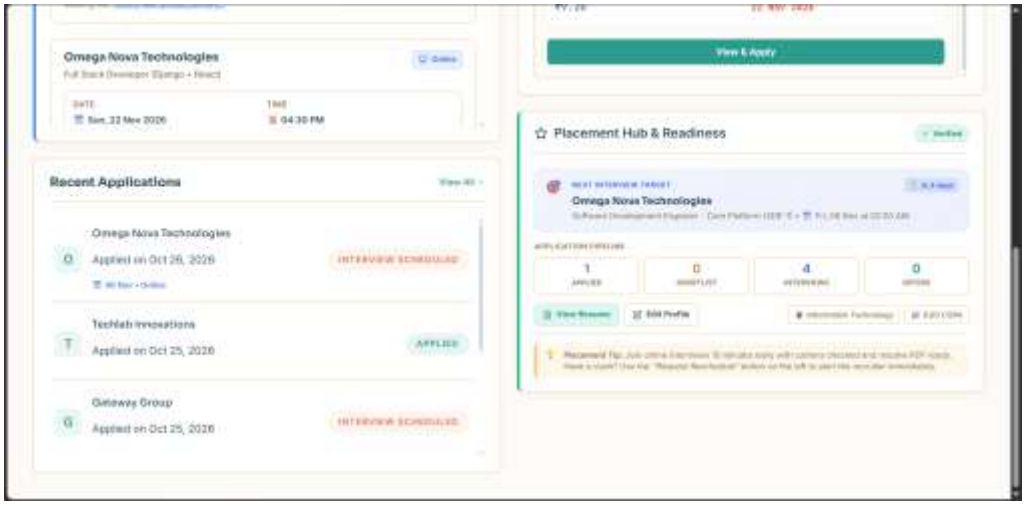


Figure 8: Student Dashboard - Active Interviews & Drive Highlights (Student Side)

Lower section of the Student Dashboard displaying curated placement drive recommendations matching the candidate's department and recent application submission statuses.

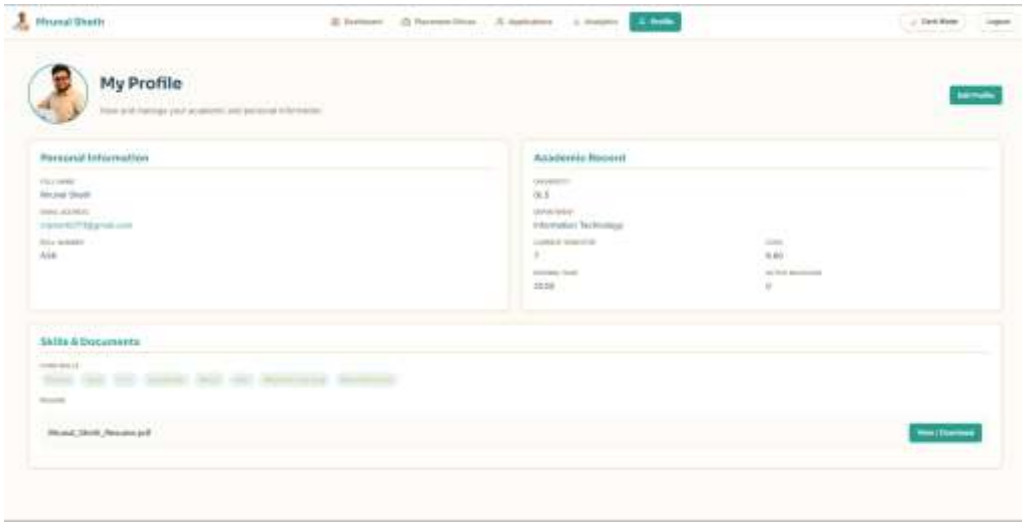


Figure 9: Student Academic Profile View (Student Side)

Detailed academic profile view presenting verified student credentials including cumulative CGPA, department affiliation, active backlogs status, tagged technical skills, and uploaded PDF resume.

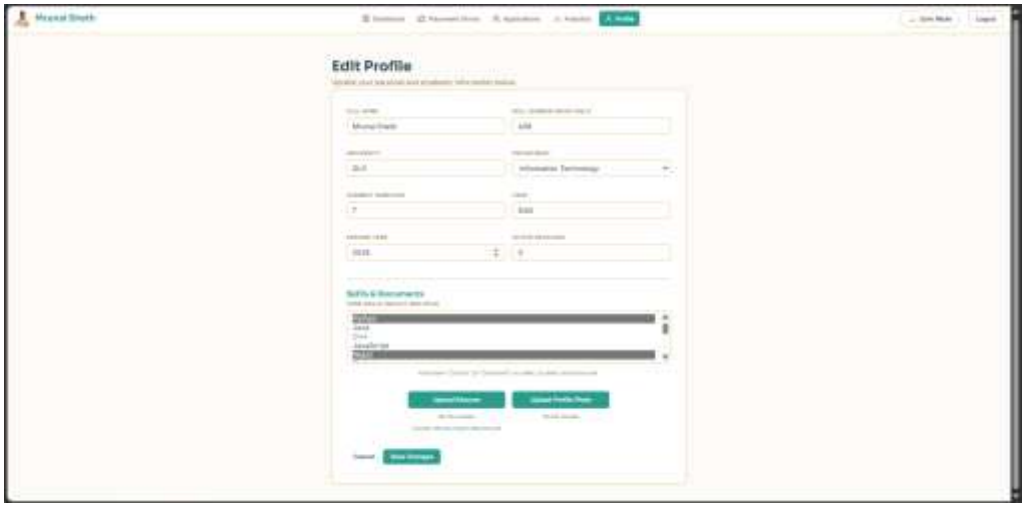


Figure 10: Edit Student Profile & Resume Management (Student Side)

Profile editing interface allowing students to maintain updated contact details, portfolio links, technical skill competencies, and upload refreshed PDF resume files.

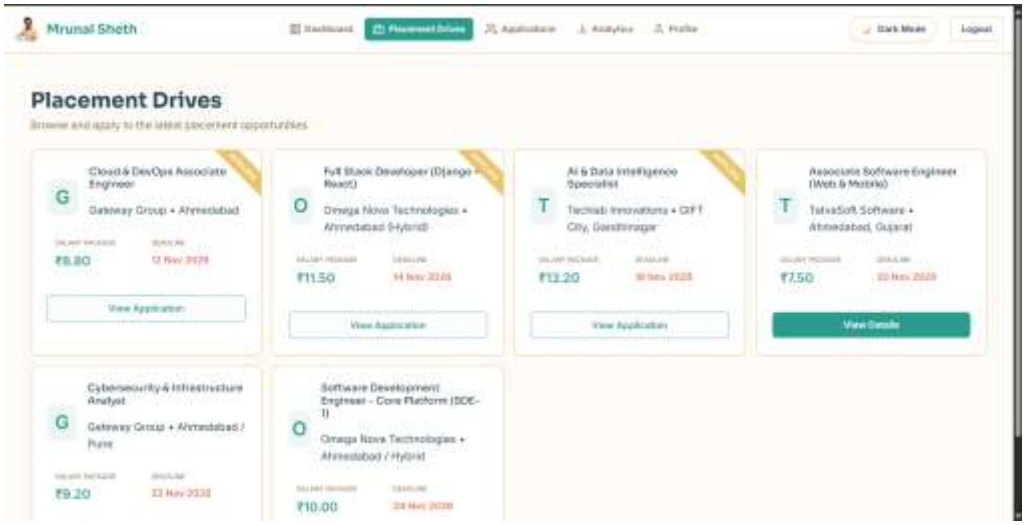


Figure 11: Campus Placement Drives Directory (Student Side)

Searchable directory of active campus placement drives featuring real-time search, package CTC badges, application deadline alerts, and instantaneous eligibility tags.



Figure 12: Placement Drive Specifications & Overview (Student Side)

Detailed placement drive overview highlighting recruiting company background, job designation, annual CTC package, job location, responsibilities, and selection round descriptions.

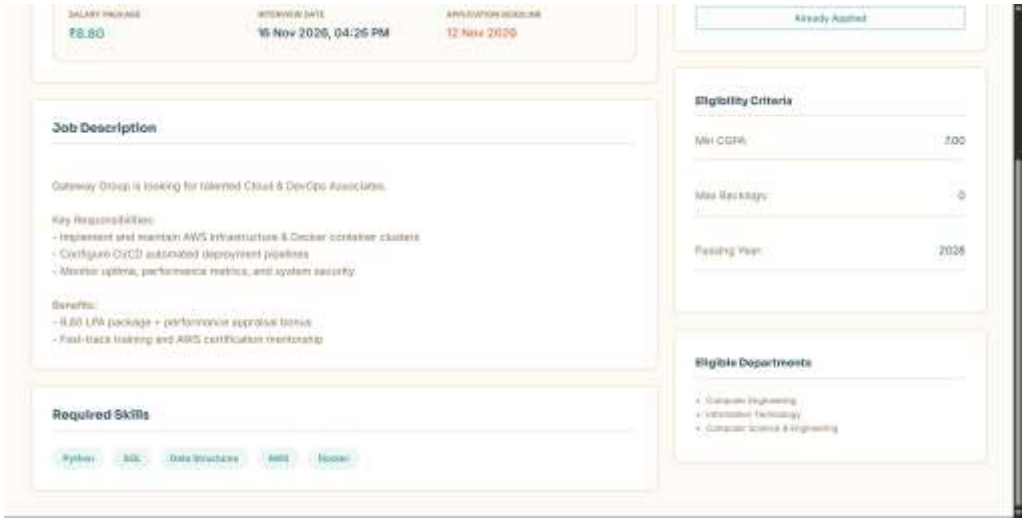


Figure 13: Drive Automated Eligibility Breakdown & Apply (Student Side)

Real-time automated eligibility evaluation panel comparing the student's academic standing against corporate cutoffs (CGPA, department, backlogs, passing year) with an enabled one-click Apply button.

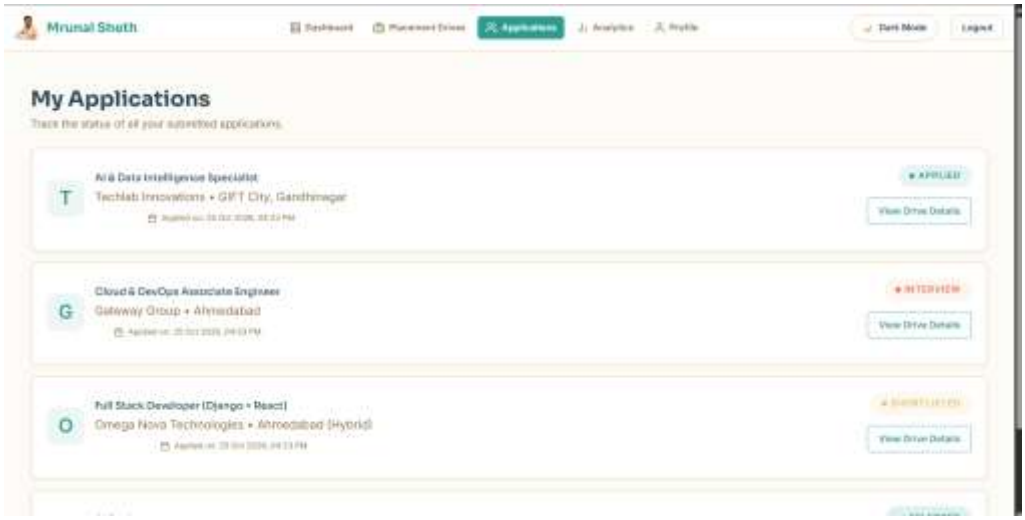


Figure 14: Student Applications Tracking & Status Board (Student Side)

Centralized application tracker where students monitor the progress of all submitted applications across stages (Applied, Shortlisted, Interview Scheduled, Selected, Rejected) with color-coded badges.

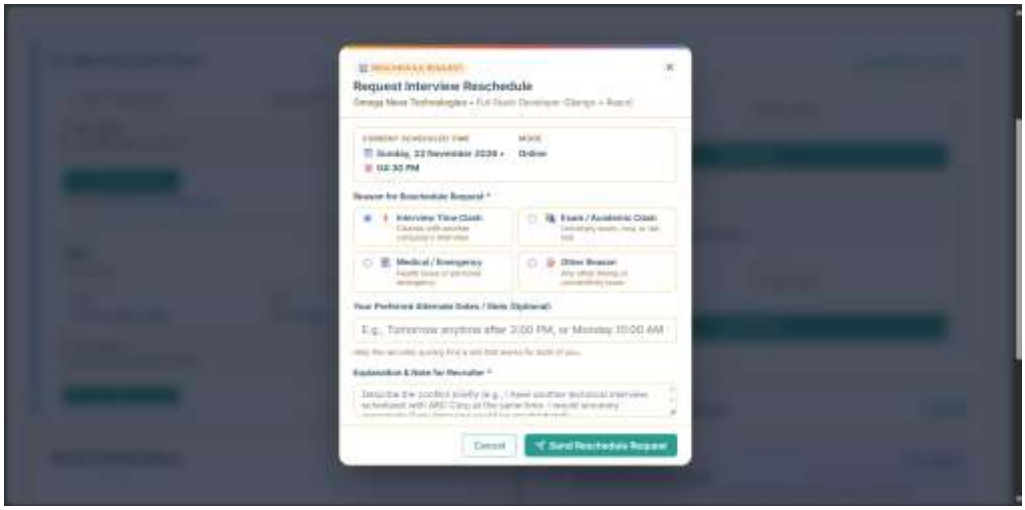


Figure 15: Student Interview Reschedule Request Modal (Student Side)

Interactive modal dialog enabling candidates to formally submit an interview reschedule request with conflict reasons (e.g., exam clash, prior commitment) and detailed explanatory notes for recruiters.

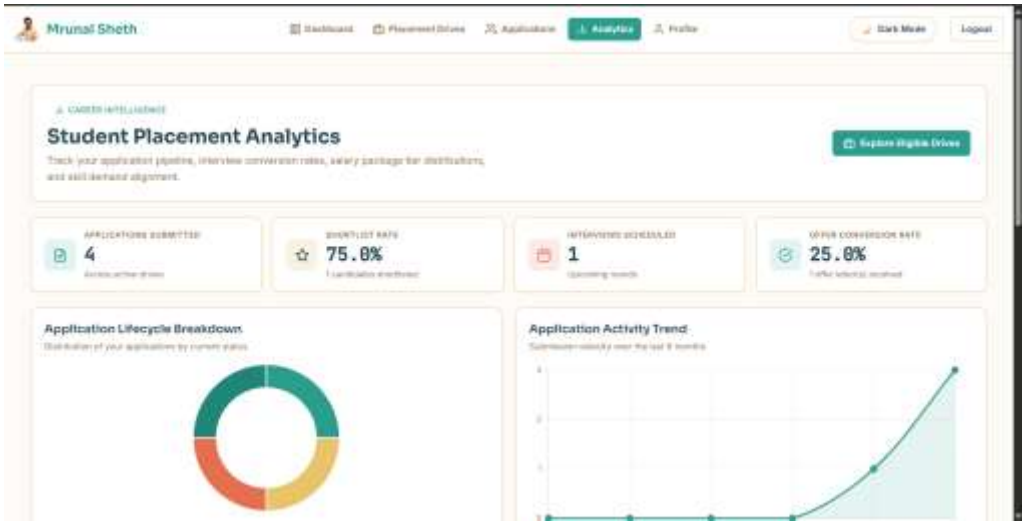


Figure 16: Student Placement Analytics - Key Performance Indicators (Student Side)

Executive placement analytics dashboard displaying overall university placement percentages, average compensation packages, and drive participation rates.



Figure 17: Student Analytics - Departmental Hiring Trends (Student Side)

Interactive Chart.js visualization presenting departmental candidate selections and hiring distributions across college branches.

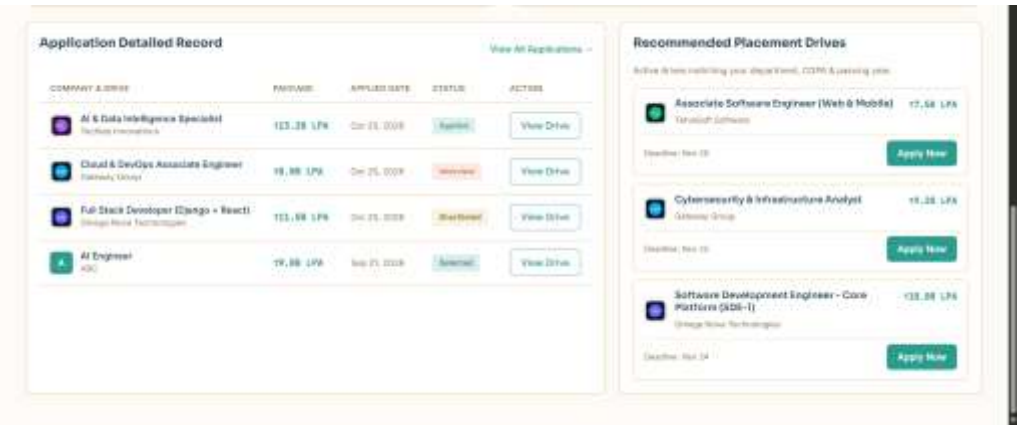


Figure 18: Student Analytics - Package & Salary Distribution (Student Side)

Salary package tier distribution chart categorizing campus placement offers across Mass, Dream, and Super Dream compensation bands.

5.3 Company Side: Recruiter & Placement Management

The Company Portal empowers corporate talent teams to manage corporate branding, configure multi-criteria placement drives, screen candidate applications, evaluate student resumes, schedule interviews with video meeting integration, manage reschedule requests, and analyze hiring conversion funnels.

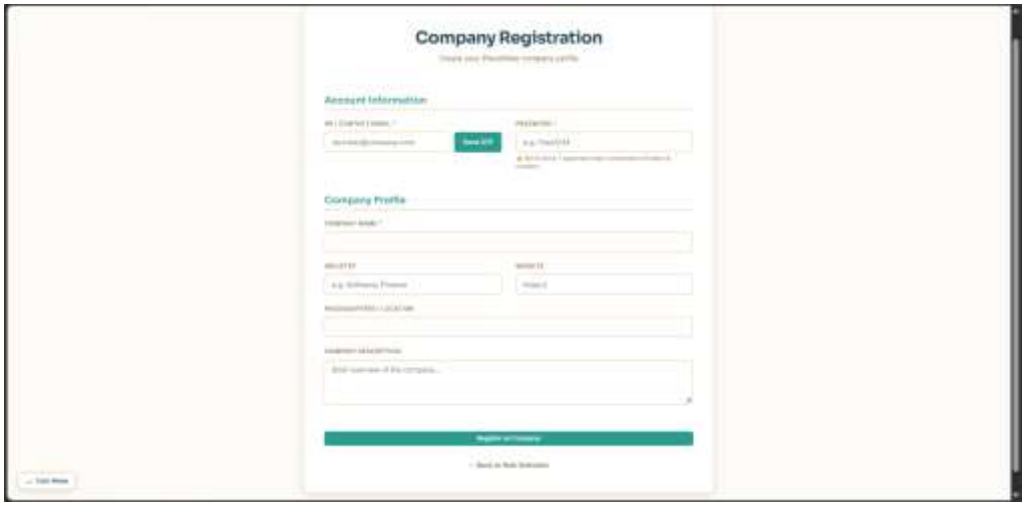


Figure 19: Company Recruiter Registration Form (Company Side)

Corporate onboarding portal enabling HR representatives to create an organizational profile with corporate email verification, industry classification, website, and headquarters location.

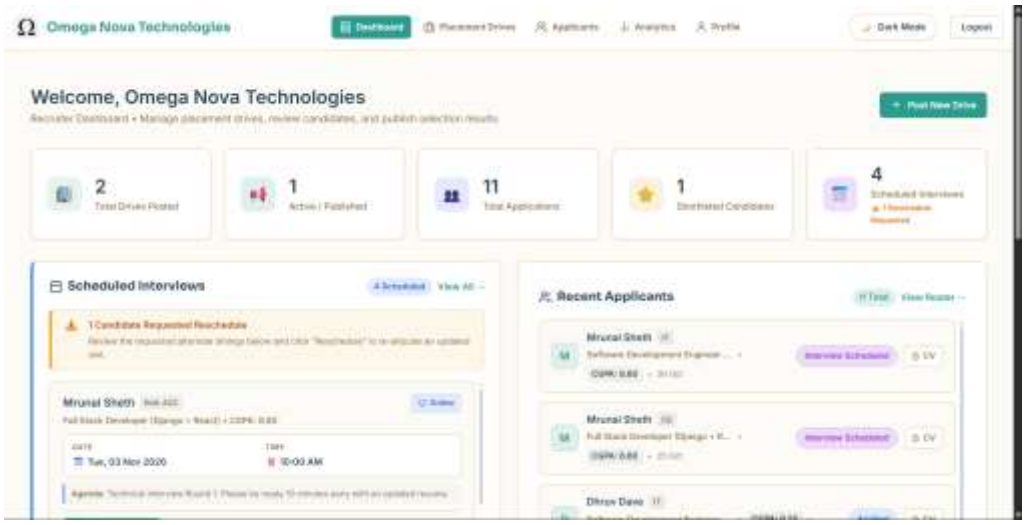


Figure 20: Company Recruiter Dashboard - Top Overview & Scheduled Interviews (Company Side)

Upper section of the balanced company dashboard featuring upcoming scheduled interviews with 'Start Meeting' video launch buttons, reschedule controls, and real-time recent applicant cards with avatar initials and CV links.

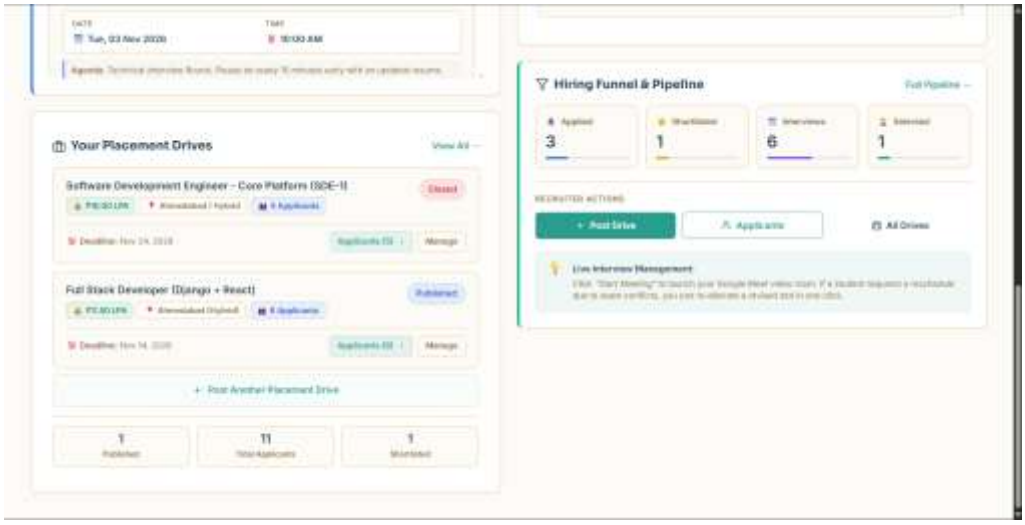


Figure 21: Company Recruiter Dashboard - Active Drives & Hiring Funnel (Company Side)

Lower section of the recruiter dashboard highlighting active placement drives, candidate distribution across the 4-stage recruitment pipeline, and recruiter launchpad action buttons.

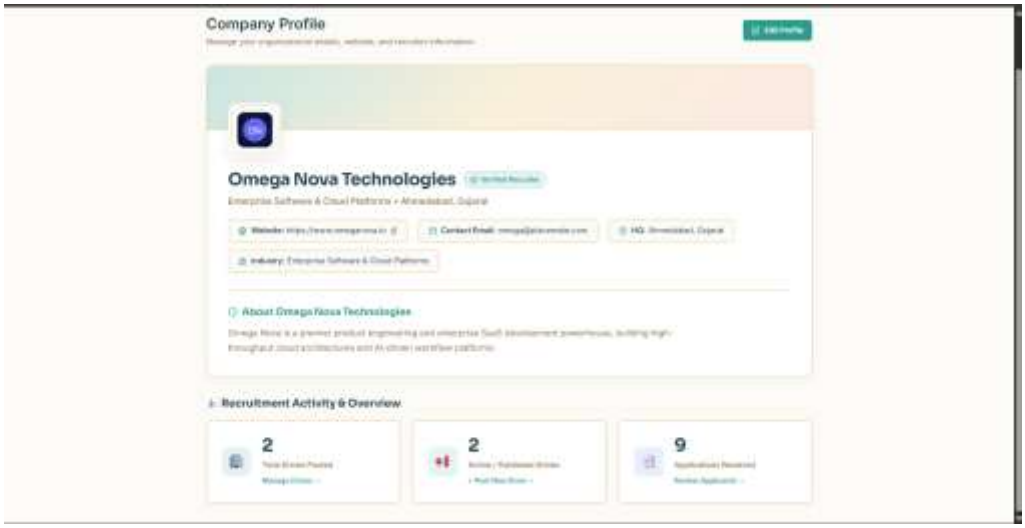


Figure 22: Company Corporate Profile View (Company Side)

Corporate profile presentation displaying organizational branding, corporate logo, industry vertical, official website, contact email, and active placement drives overview.

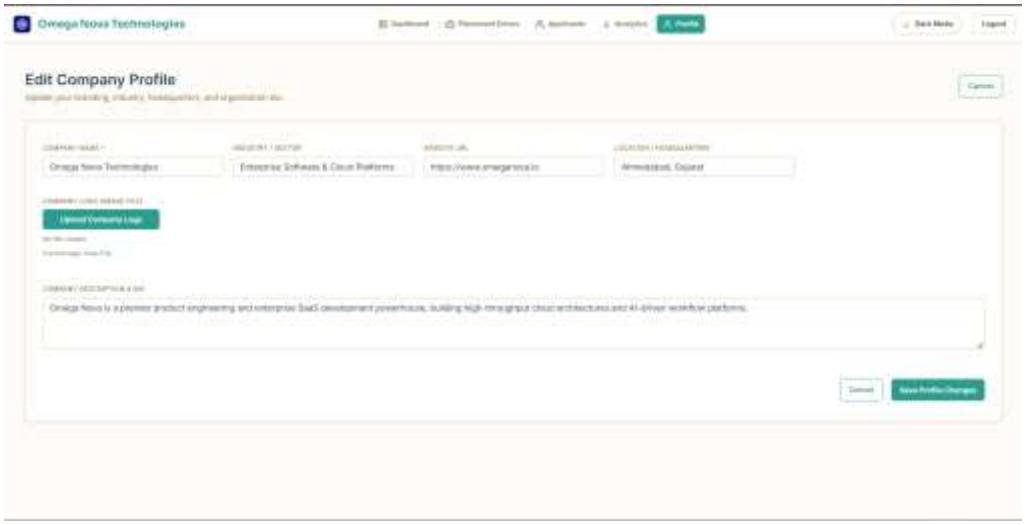


Figure 23: Edit Company Profile & Branding (Company Side)

Company profile editing form allowing HR personnel to update recruiter contact numbers, organizational summaries, corporate logos, and recruitment guidelines.

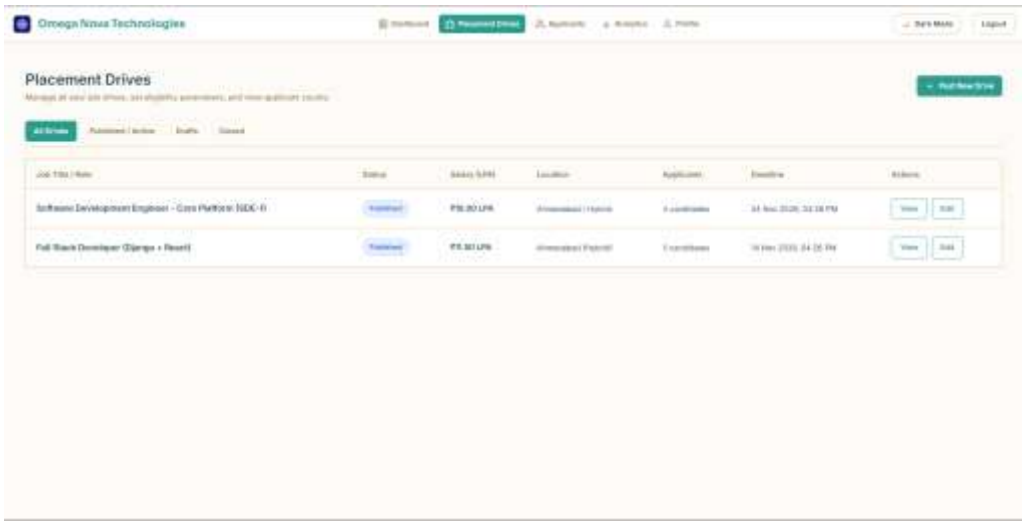


Figure 24: Company Placement Drives Management (Company Side)

Overview management table detailing all placement drives published by the organization, showing CTC packages, applicant counts, deadlines, and drive lifecycle statuses.

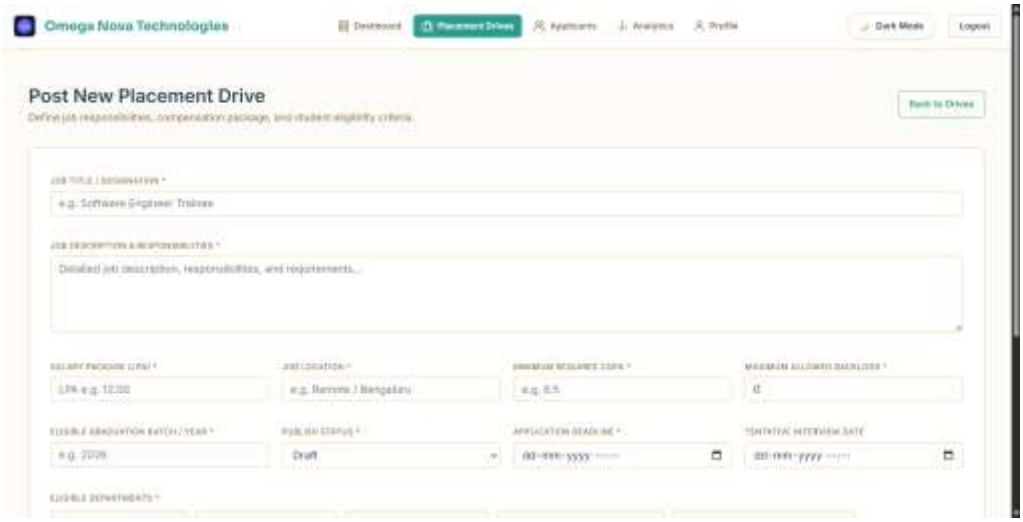


Figure 25: Create Placement Drive - Job Role & Package Details (Company Side)

Step 1 of the drive creation wizard where recruiters define job designation, annual CTC compensation package, job location, detailed job description, and application closing date.

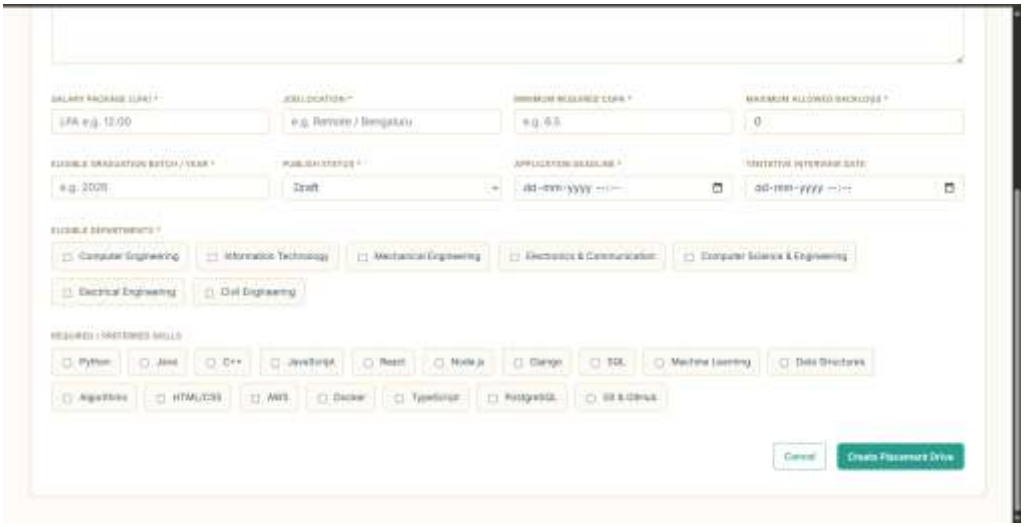


Figure 26: Create Placement Drive - Academic Eligibility Cutoffs (Company Side)

Step 2 of the drive creation wizard where recruiters establish strict filtering criteria: minimum CGPA cutoff, maximum allowed backlogs, eligible passing year, and eligible engineering departments.

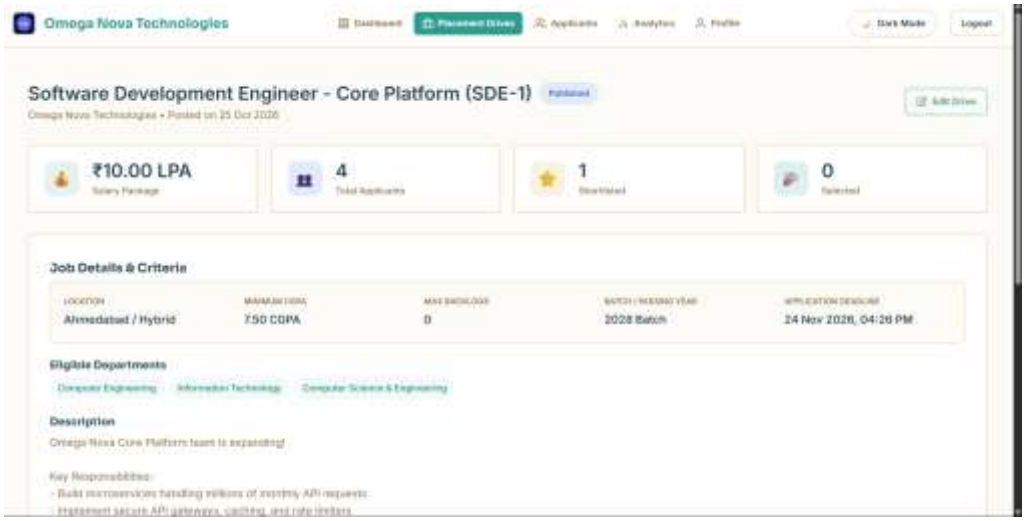


Figure 27: Drive Details & Applicant Funnel Metrics (Company Side)

Analytical drive overview illustrating key recruitment milestones: total applicants, shortlisted candidates, scheduled interviews, and final accepted offers.

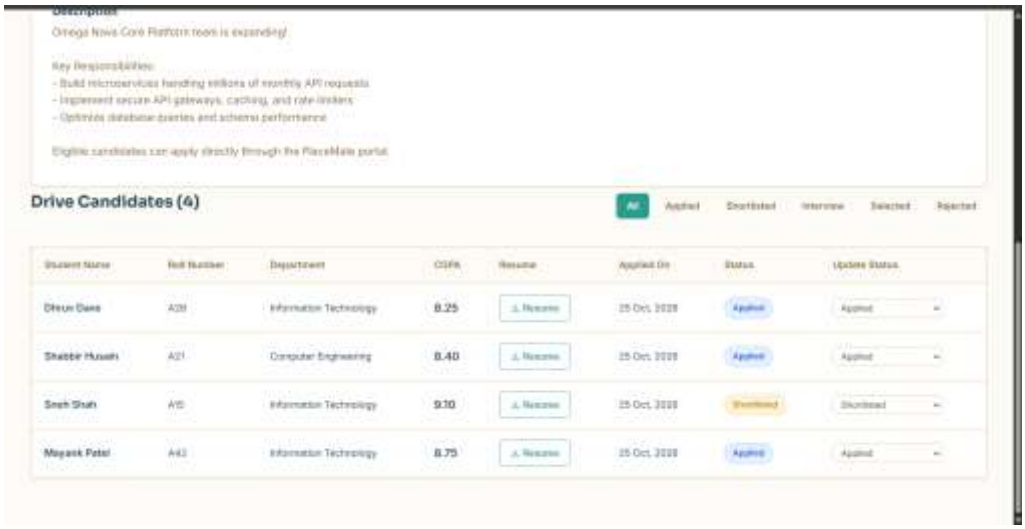


Figure 28: Drive Requirements & Candidate Screening Controls (Company Side)

Drive configuration overview displaying eligibility requirements, quick applicant filter shortcuts, and drive status toggle controls.

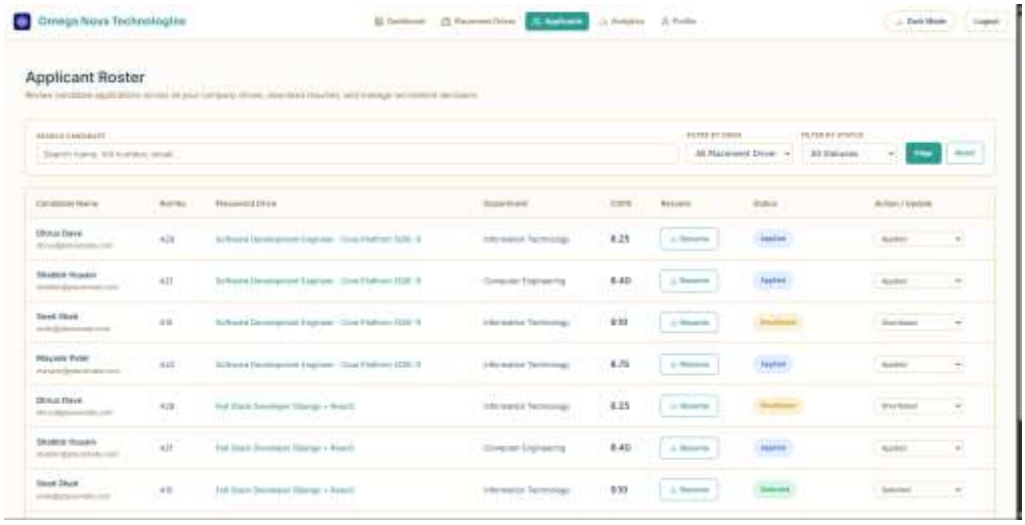


Figure 29: Drive Applicants Pipeline Overview (Company Side)

Dedicated applicant screening dashboard featuring search, departmental filters, candidate recruitment status, and instant resume download capabilities.

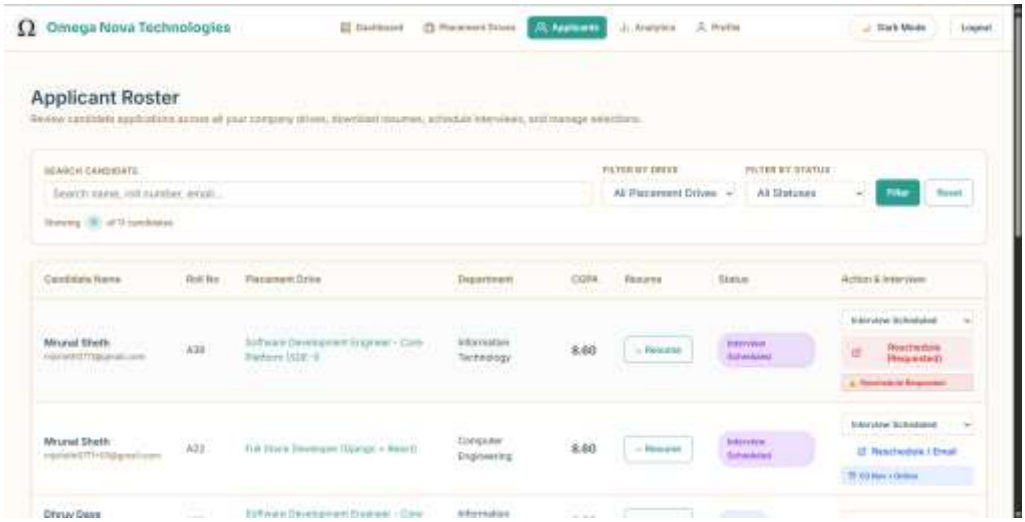


Figure 30: Candidate Academic Profile & Resume Evaluation (Company Side)

Comprehensive candidate evaluation card displaying academic credentials, verified CGPA, semester, technical skills, and resume preview for recruiters.

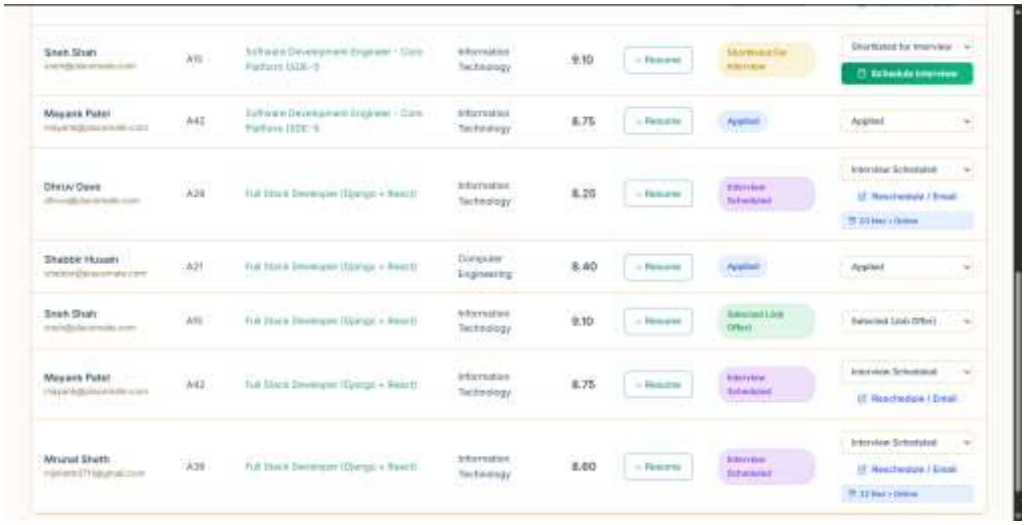


Figure 31: Candidate Recruitment Pipeline Stage Management (Company Side)

Dynamic candidate management view enabling recruiters to transition applicants between recruitment stages: Shortlisted, Schedule Interview, Selected, and Rejected.

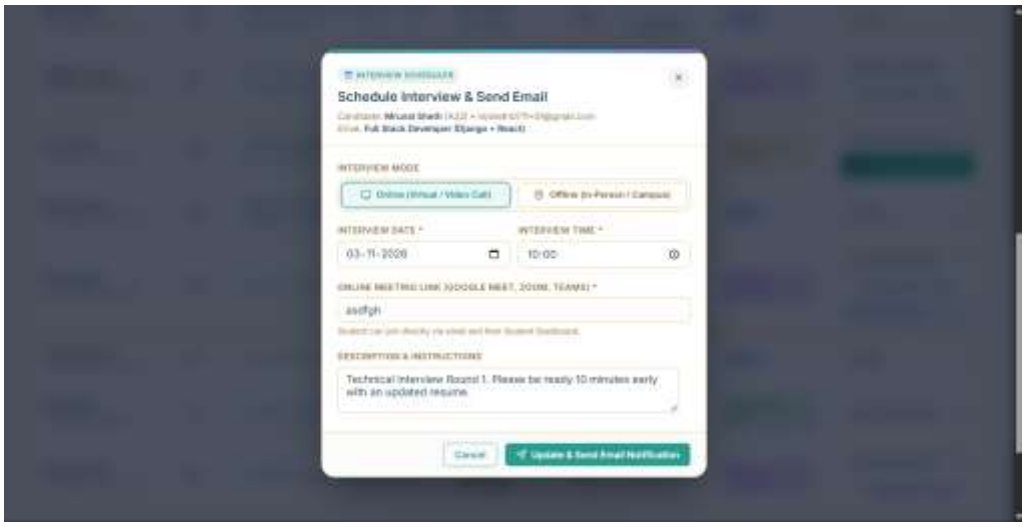


Figure 32: Schedule Interview Modal - Mode, Date & Meeting Link (Company Side)

Interview scheduling modal allowing recruiters to configure interview format (Virtual Video Call vs In-Person), select date and time, and embed video links (Google Meet, Teams, Zoom).

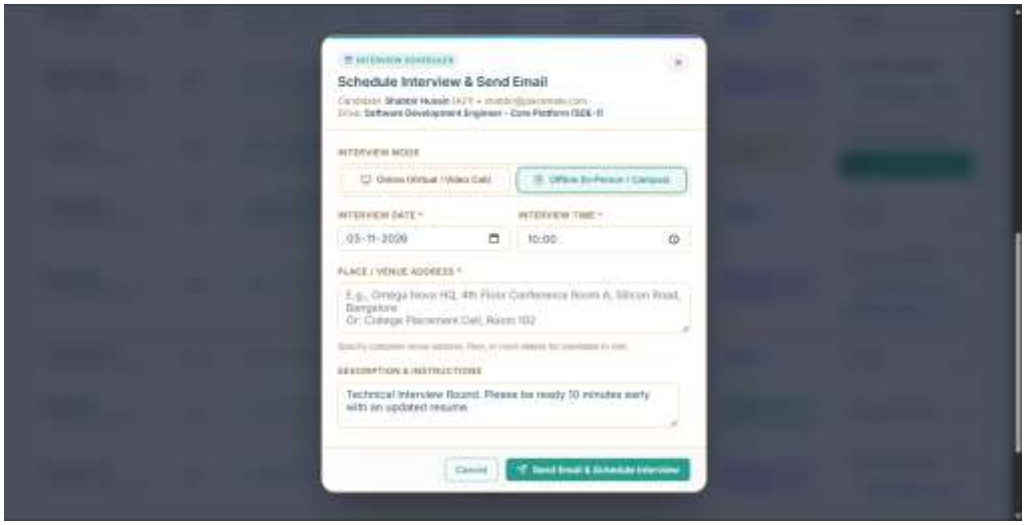


Figure 33: Schedule Interview Modal - Instructions & Calendar Setup (Company Side)

Interview configuration panel for inputting candidate instructions, technical interview guidelines, and dispatching automated calendar invites with confirmation emails.

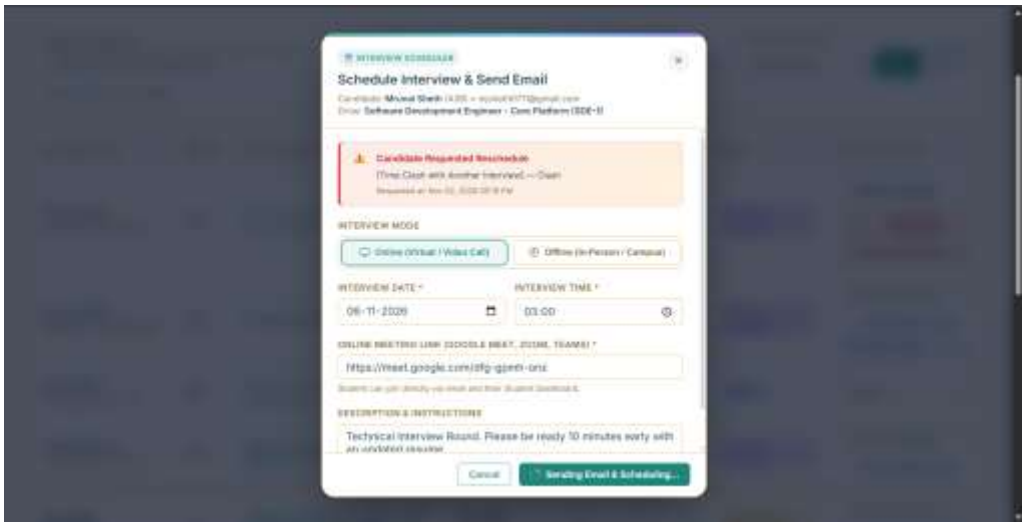


Figure 34: Company Interview Reschedule Review & Action Modal (Company Side)

Scrollable interview reschedule dialog highlighting candidate conflict reasons and enabling recruiters to select an alternate interview date and time slot.

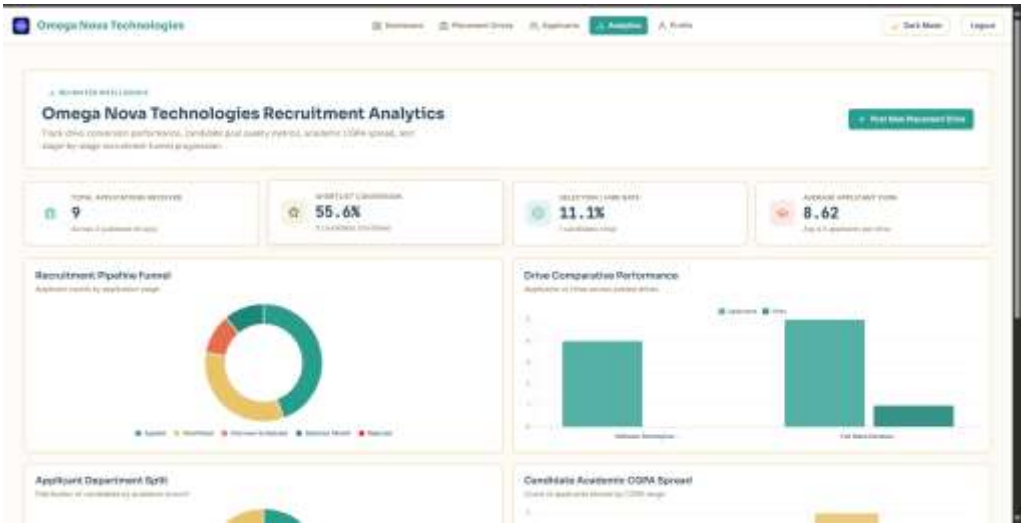


Figure 35: Company Recruitment Analytics - Conversion Funnel (Company Side)

Recruitment analytics screen presenting the company's application-to-selection hiring funnel and candidate conversion rates.

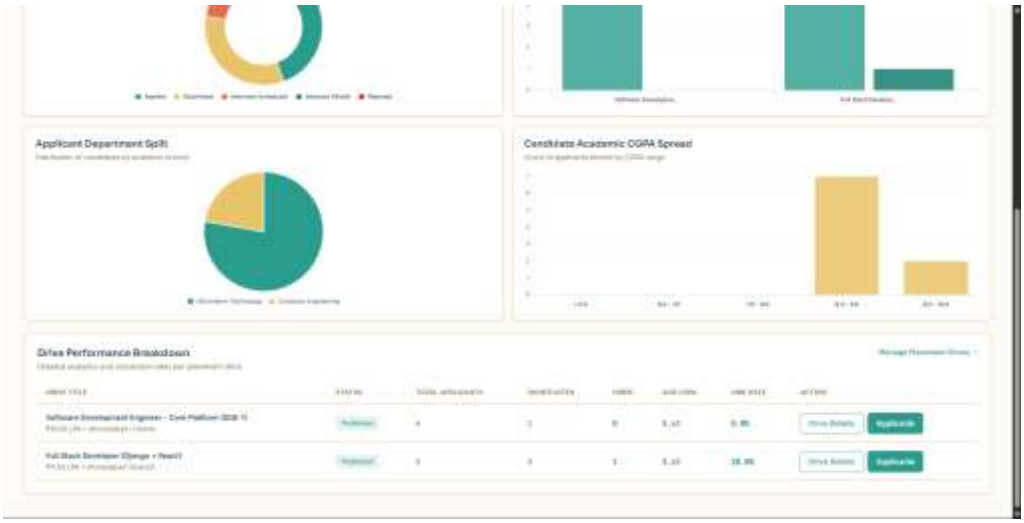


Figure 36: Company Analytics - Departmental Applicant Demographics (Company Side)

Visual demographic breakdown illustrating applicant distributions across engineering disciplines to evaluate talent pool diversity.

5.4 Admin Side: Placement Officer Administration

The Placement Officer (Admin) Portal equips institutional placement cell heads with centralized governance tools to oversee registered users, verify corporate accounts, maintain academic master catalogs, approve placement drives, and monitor global application logs.

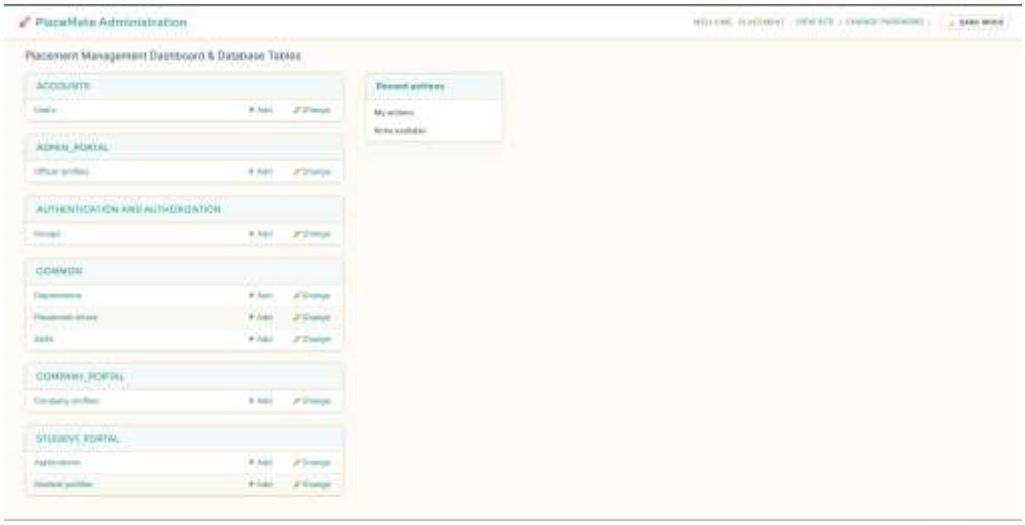


Figure 37: Placement Officer Admin Dashboard (Admin Side)

Central administration dashboard delivering university-wide recruitment KPIs: total registered students, verified companies, active placement drives, and cumulative placements.

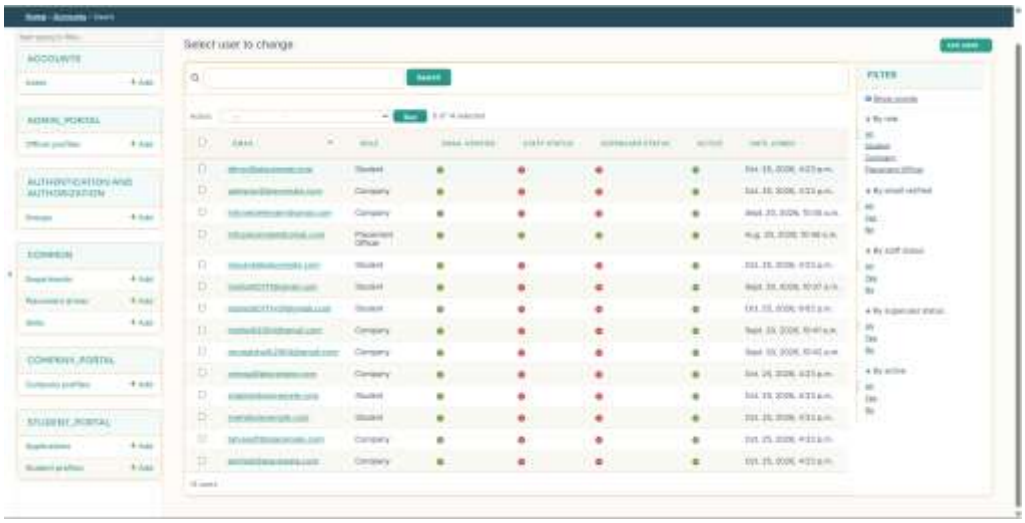


Figure 38: Central User Management & Role Verification (Admin Side)

Central administrative directory for inspecting, verifying, and managing student, recruiter, and coordinator accounts with granular role permissions.

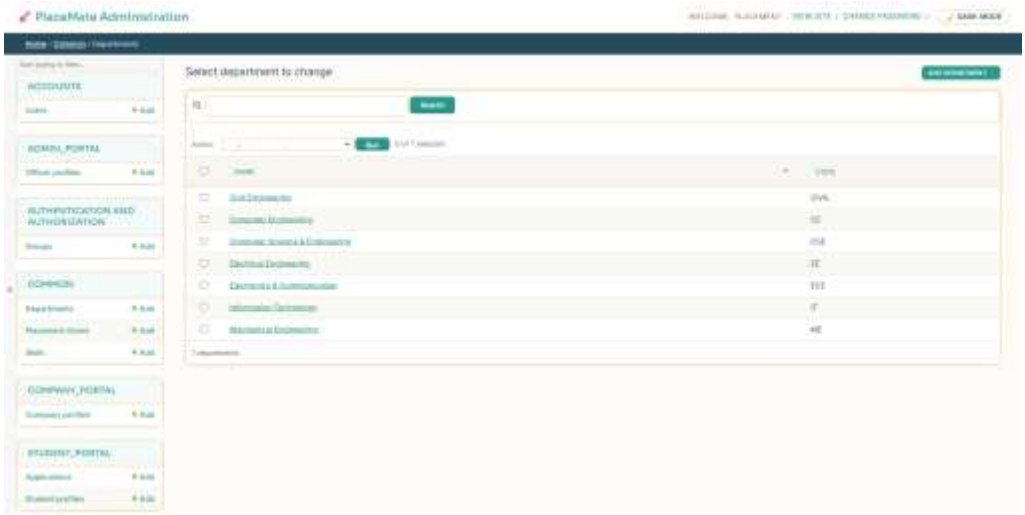


Figure 39: Academic Department Master Management (Admin Side)

Master data management interface allowing placement officers to create, edit, and organize academic departments and degree program codes.

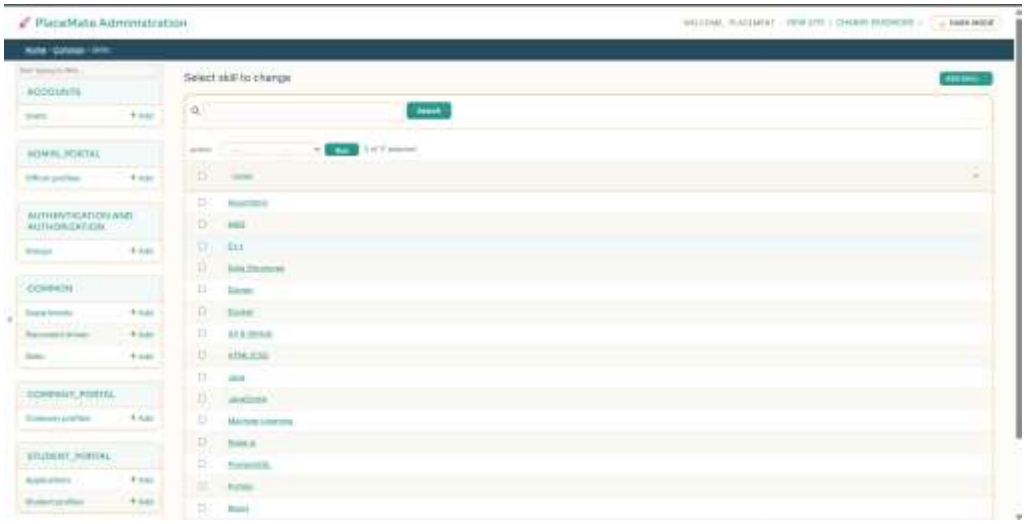


Figure 40: Technical Skills Master Directory (Admin Side)

Master repository of technical skills and professional competencies utilized across candidate profiles and corporate drive eligibility matching.

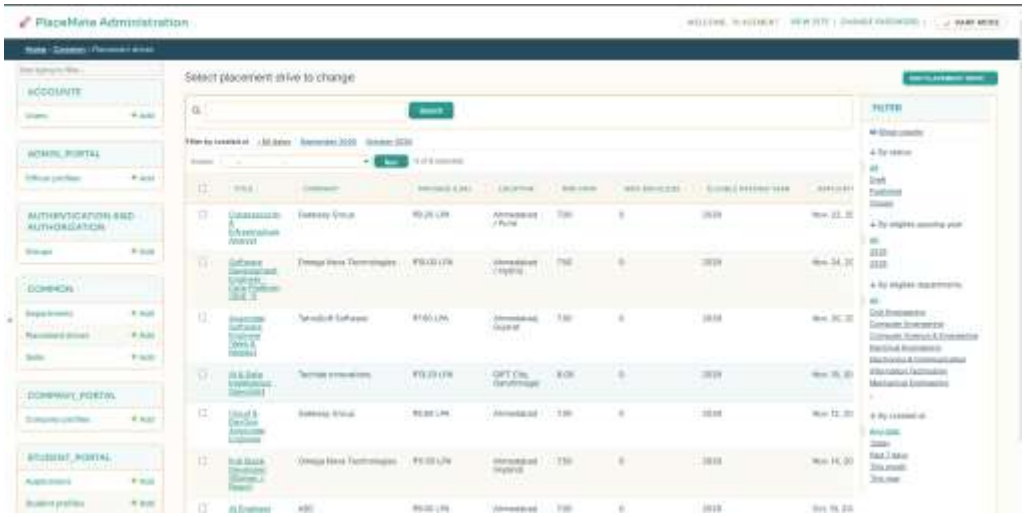


Figure 41: Campus Placement Drives Oversight & Moderation (Admin Side)

Institutional oversight panel for monitoring drives across all visiting recruiters, validating compensation offerings, and approving drives for student visibility.

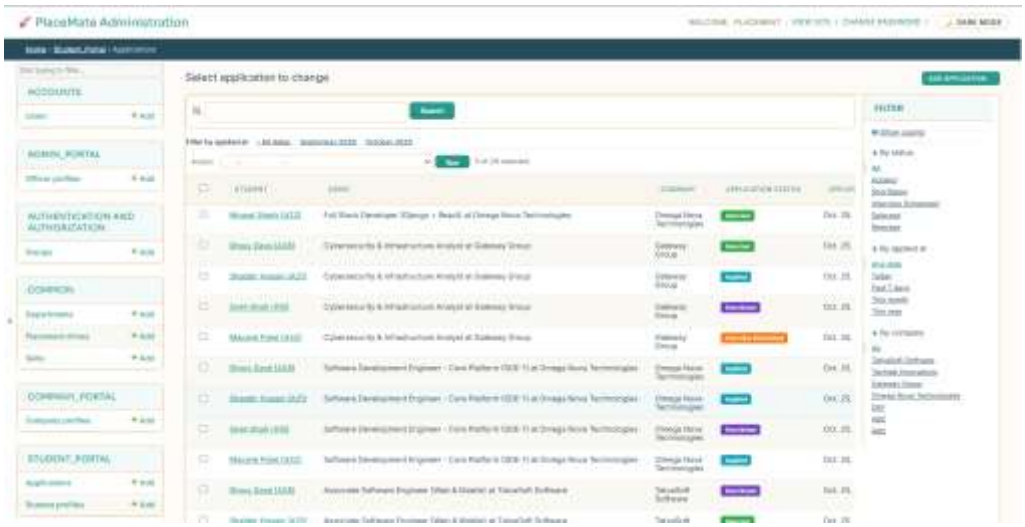


Figure 42: Global Job Application Records Oversight (Admin Side)

Comprehensive institutional audit table logging all student job applications across campus drives with complete timestamp history and current recruitment statuses.

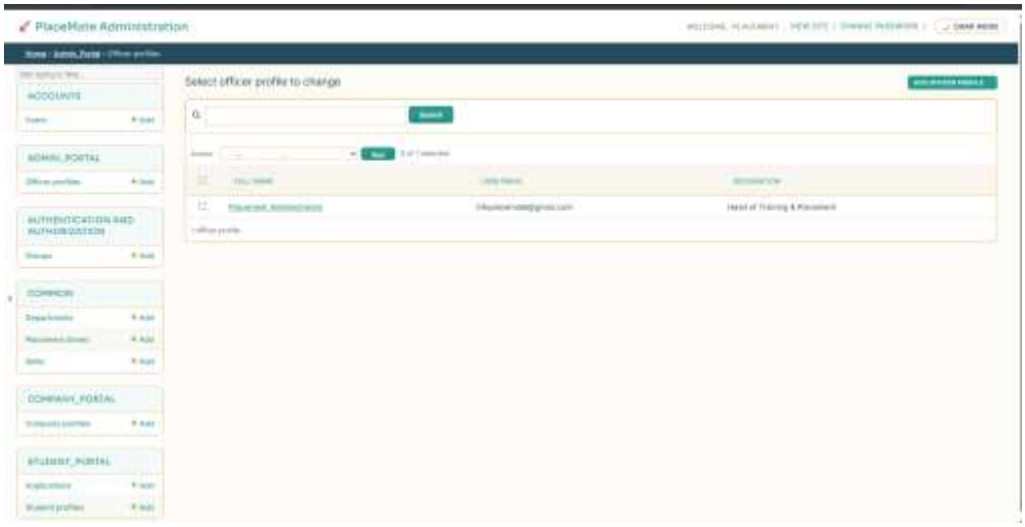


Figure 43: Placement Officer Staff Profiles & Admin Directory (Admin Side)

Administrative roster managing placement cell faculty coordinators, staff profiles, and administrative access privileges.

5.5 Automated Email Notification System

PlaceMate integrates an automated email dispatch engine engineered with responsive, branded HTML templates. Whenever critical milestones occur, the server generates and dispatches immediate notification emails directly to candidate and recruiter inboxes.

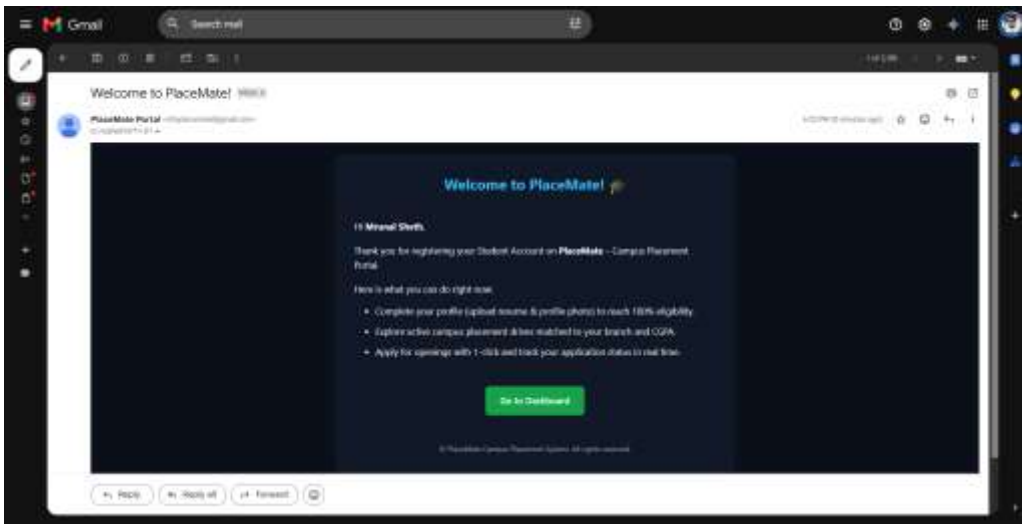


Figure 44: Welcome & Account Verification Email (Mails)

Responsive HTML onboarding email delivered upon successful user verification, confirming account details and providing direct portal access links.

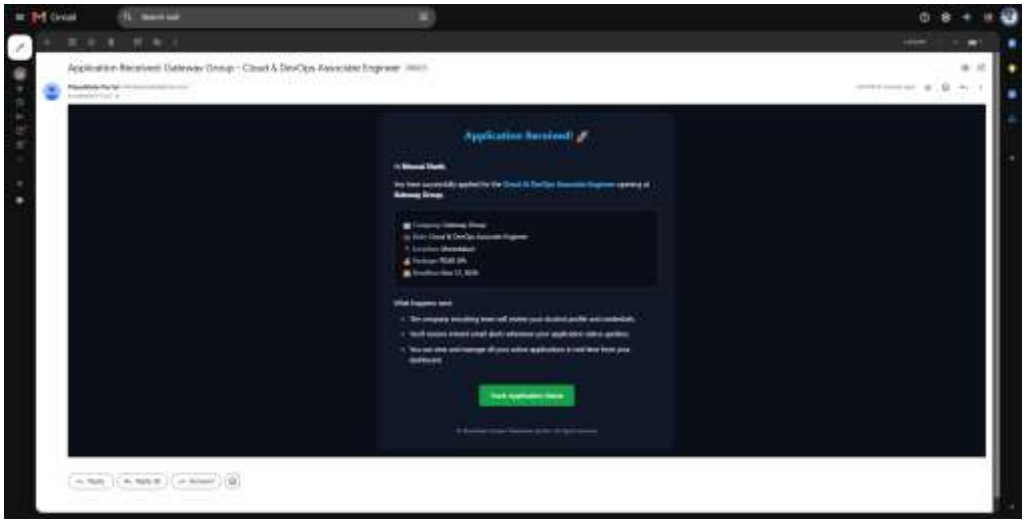


Figure 45: Placement Drive Application Confirmation Email (Mails)

Automated application receipt dispatched to students immediately upon applying for a campus drive, detailing job role, company name, and application reference ID.

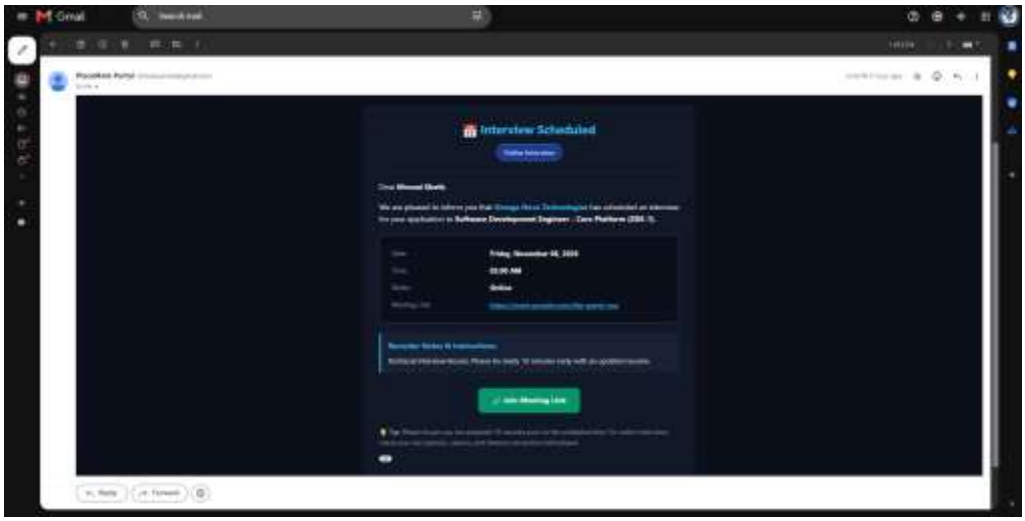


Figure 46: Interview Invitation & Schedule Notification Email (Mails)

Comprehensive interview invitation email detailing date, time slot, meeting mode, video call link, and round preparation instructions.

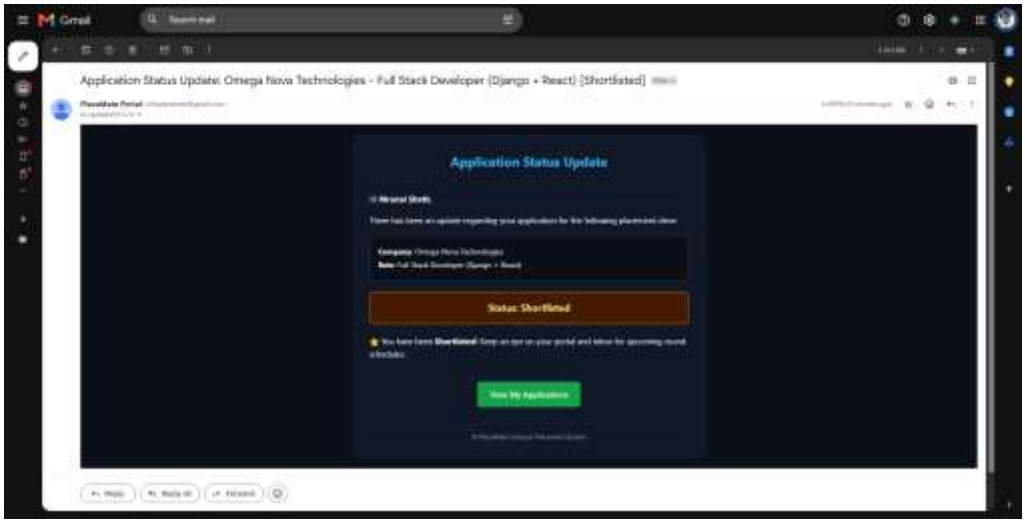


Figure 47: Candidate Shortlisted Notification Email (Mails)

Instant status alert informing the student of successful candidate shortlisting and outlining upcoming assessment and interview expectations.

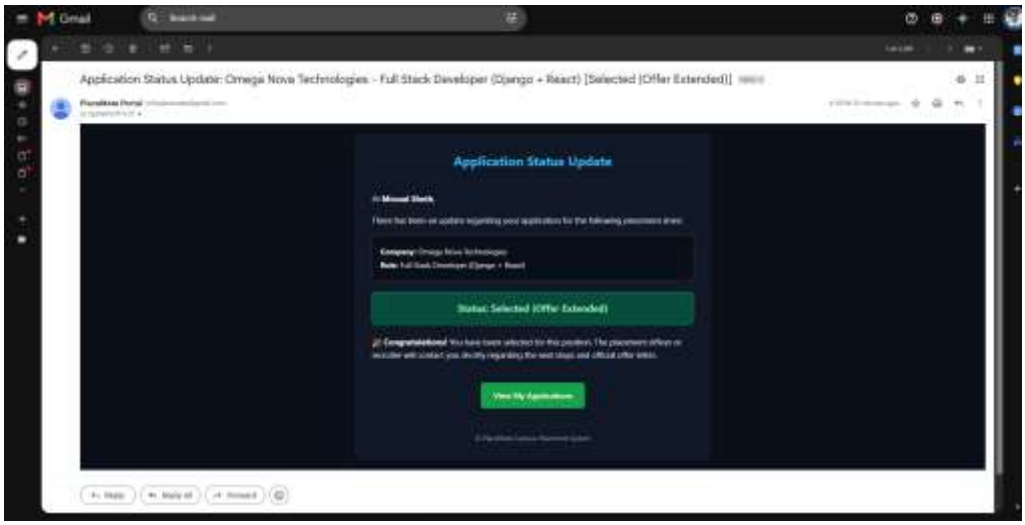


Figure 48: Final Selection & Job Offer Congratulatory Email (Mails)

Formal congratulatory offer notification sent to selected candidates with compensation package (CTC) and onboarding instructions.

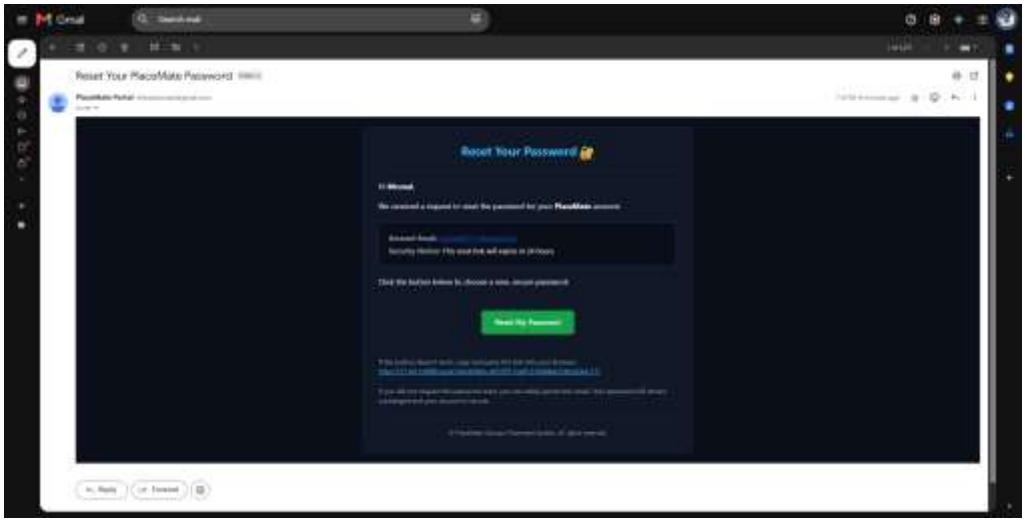


Figure 49: Secure Password Reset Link Request Email (Mails)

Time-sensitive password reset email delivering a cryptographically secured one-time token link to restore account access.

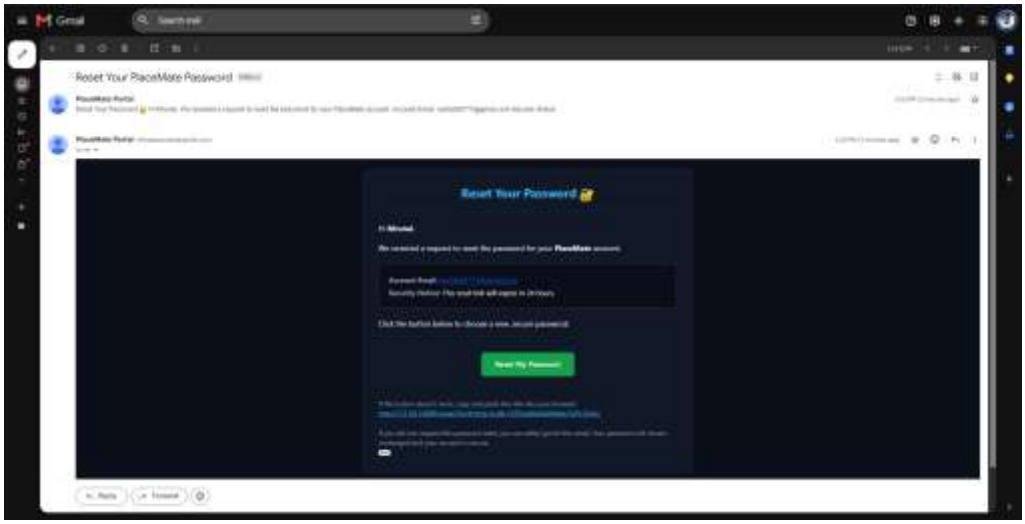


Figure 50: Password Successfully Reset Confirmation Email (Mails)

Automated account security notification alerting the user that their account password was successfully updated.

6. Conclusion

PlaceMate successfully modernizes the traditional, fragmented campus recruitment ecosystem into an integrated, automated, and secure digital platform. By uniting Students, Corporate Recruiters, and College Placement Officers within a unified architecture, the system eliminates manual administrative verification, prevents human errors in cutoff eligibility checking, and ensures transparent communication through real-time application tracking and automated emails.

The application's responsive interface, zero-flicker light and dark mode toggling, interactive recruitment analytics, and robust interview scheduling engine deliver a polished, enterprise-ready experience for all university stakeholders. PlaceMate provides an extensible, scalable, and production-tested foundation that modernizes institutional recruitment and empowers colleges to conduct organized, efficient, and data-driven placement seasons.